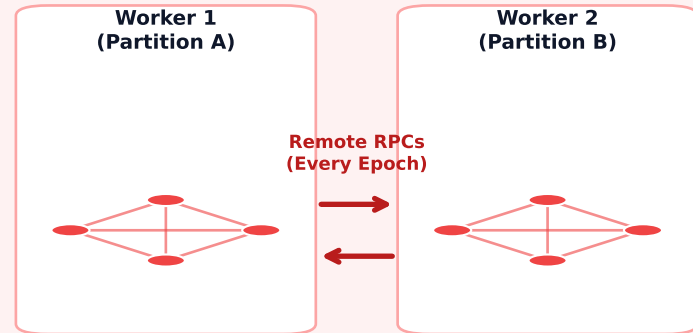
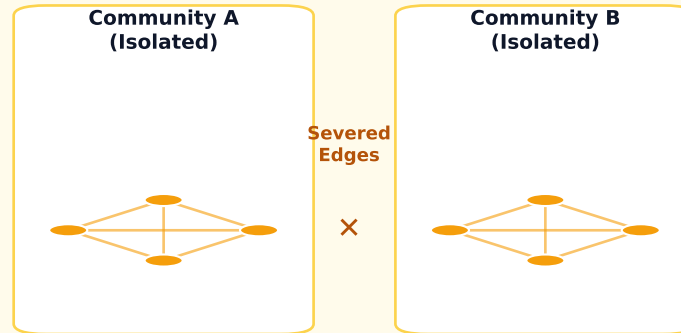


(a) Synchronized Distributed GRL



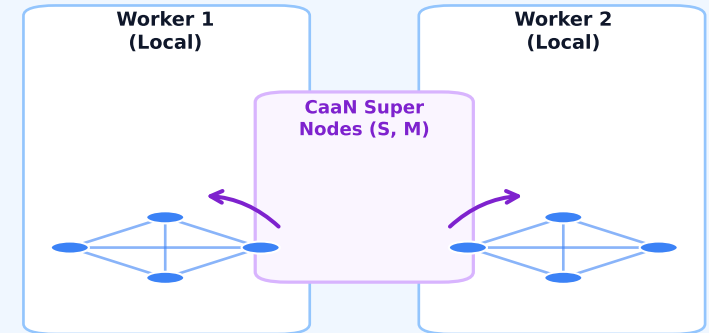
- Full-graph state split across cluster
- Heavy inter-worker RPC sync (28-46 GB)
- High network communication overhead
- Costly GPU / parameter server clusters

(b) Naive Community Partitioning



- Isolated training on subgraphs
- 0.0 GB inter-worker network traffic
- Boundary nodes lose external context
- Severe accuracy drop (95% $\rightarrow$ 74%)

(c) EMO: Relational CaaN Solution



- Transactional Delta Lake on Amazon S3
- 0.0 GB training network traffic
- Relational CaaN super-node broadcast
- Restores 98%+ full-graph accuracy